

TRUNG TUAN MAI

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EDUCATION

Bachelor of Science, Software Engineering

Expected April 2027

Schulich School of Engineering, University of Calgary

CORE SKILLS

Programming: Python, Java, JavaScript, C/C++

Frontend: HTML, CSS, JavaScript

Backend: Flask, Spring Boot, REST APIs

Databases: MongoDB, MySQL, H2

Tools: Git, Docker, Gradle, Linux

Concepts: full-stack development, API design, database systems, event-driven architecture

PROJECTS

Movie Ticket Reservation System

(Software Design Course)

April 2025

- Developed a Java-based reservation system using JDBC and MySQL to manage bookings, seats, and transactions, enabling consistent end-to-end reservation workflows.
- Implemented booking and cancellation logic, ensuring synchronized updates between application state and database persistence.
- Optimized relational queries to improve transaction consistency and reduce data synchronization issues.

Predicting Car Sales Revenue – Machine Learning System

(Machine Learning Course)

December 2025

- Developed a machine learning pipeline in Python using pandas, NumPy, and scikit-learn to process a dataset exceeding **500K records**.
- Engineered predictive features and evaluated regression models, achieving **$R^2 = 0.91$** for accurate forecasting.

Data Ingestion & Backend System

(Web-Based Systems Course)

February 2026

- Engineered a Flask backend with REST APIs to ingest and normalize multi-source datasets into MongoDB, enabling consistent data storage across heterogeneous inputs.
- Implemented schema validation and unique-key deduplication to enforce data integrity and prevent record conflicts.
- Designed an ETL-style pipeline combining scraping, parsing, transformation, and persistence to automate data processing workflows.
- Diagnosed and resolved API-database integration issues to improve system reliability under concurrent execution.

Backend Job-Market Simulation System

(Software Architecture Course)

February 2026

- Engineered a Flask and SQLAlchemy backend managing player progression and application state in a relational database.
- Designed REST APIs for lifecycle management, activity tracking, and application history retrieval, supporting client-server state synchronization.
- Implemented rule-based business logic to compute outcomes based on system constraints and progression rules.

EXTRA-CURRICULARS

JP Morgan Chase Virtual - Financial Transaction Processing Backend

January 2026

- Developed Spring Boot backend services exposing REST endpoints for account persistence and balance retrieval.
- Integrated Kafka-based message listeners to simulate asynchronous transaction processing, improving system scalability and decoupling service components.

Calgary Hackathon — Godot Arena System

February 2026

- Collaborated with a cross-functional team to design and deliver a working prototype under tight deadlines.